

Homework #1 - Solutions

Companies(cid: integer, *cname*: string, *address*: string)

Products(pid: integer, *pname*: string)

Produce(cid: integer, pid: integer, *cost*: real)

1. Write the following queries in relational algebra:

a) Find the *names* of companies who produce products with *pname*="milk" .

$$\prod_{cname} ((\sigma_{pname="milk"}(PRODUCTS)) \bowtie PRODUCE \bowtie COMPANIES)$$

b) Find the *cids* of companies who produce "milk" or "sugar".

$$\prod_{cid} ((\sigma_{pname="milk" \vee pname="sugar"}(PRODUCTS)) \bowtie PRODUCE)$$

c) Find the *cids* of companies who produce both "milk" and "sugar".

$$\begin{aligned} & \prod_{cid} ((\sigma_{pname="milk"}(PRODUCTS)) \bowtie PRODUCE) \\ & \cap \prod_{cid} ((\sigma_{pname="sugar"}(PRODUCTS)) \bowtie PRODUCE) \end{aligned}$$

d) Find the *cnames* of companies who produce some products that are produced by the company having name "XYZ".

$$\begin{aligned} & \prod_{cname} \left(\left\{ \prod_{cid} \left[\left(\prod_{pid} ((\sigma_{cname="XYZ"}(COMPANIES)) \bowtie PRODUCE) \right) \bowtie PRODUCE \right] \right. \right. \\ & \quad \left. \left. - \prod_{cid} (\sigma_{cname="XYZ"}(COMPANIES)) \right\} \bowtie COMPANIES \right) \end{aligned}$$

e) Find the *cids* of companies who produce every products.

$$\prod_{cid, pid} (PRODUCE) \quad \div \quad \prod_{pid} (PRODUCTS)$$

f) Find the *cids* of companies who produce every “milk” products.

$$\prod_{cid,pid} (PRODUCE) \div \prod_{pid} (\sigma_{pname="milk"}(PRODUCTS))$$

g) Find the *cids* of companies who produce “milk” but not “sugar”.

$$\begin{aligned} & \prod_{cid} ((\sigma_{pname="milk"}(PRODUCTS)) \bowtie PRODUCE) \\ & - \prod_{cid} ((\sigma_{pname="sugar"}(PRODUCTS)) \bowtie PRODUCE) \end{aligned}$$

h) Find the names of companies who do not produce any product.

$$\prod_{cname} ((\prod_{cid} (COMPANIES) - \prod_{cid} (PRODUCE)) \bowtie COMPANIES)$$

i) Find the names of companies who produce at least one product.

$$\prod_{cname} (\prod_{cid} (PRODUCE) \bowtie COMPANIES)$$

j) Find the name of the most expensive product.

To solve this question you need 2 copies of PRODUCE table, namely PRODUCE1 and PRODUCE 2.

$$\begin{aligned} & \prod_{pname} ((\prod_{\substack{PRODUCE1.pid, \\ PRODUCE2.pid}} (\prod_{pid,cost} (PRODUCE1) \bowtie_{PRODUCE1.cost \geq PRODUCE2.cost} \prod_{pid,cost} (PRODUCE2)) \\ & \div \prod_{pid} (PRODUCTS)) \bowtie PRODUCTS) \end{aligned}$$

k) Find the *pids* of the most expensive product produced by company named “XYZ”.

Let X be a table which has the pid and cost of products that are produced by the company XYZ, so this table is computed as follows:

$$X \leftarrow \prod_{pid,cost} ((\sigma_{cname="XYZ"}(COMPANIES)) \bowtie PRODUCE)$$

Then, take 2 copies of X that are named as X1 and X2. The result is computed as follows:

$$\left(\prod_{X1.pid, X2.pid} (X1 \bowtie_{X1.cost \geq X2.cost} X2) \right) \div \prod_{pid} (X)$$

2. Write the queries given in a, b, c, d, g, h, i in standard SQL.

a) Find the *names* of companies who produce products with *pname*="milk" .

```
SELECT C.cname
FROM COMPANIES C, PRODUCTS P, PRODUCE PR
WHERE P.pid = PR.pid AND
      PR.cid = C.cid AND
      P.pname = "milk";
```

b) Find the *cids* of companies who produce "milk" or "sugar".

```
SELECT PR.cid
FROM PRODUCTS P, PRODUCE PR
WHERE P.pid = PR.pid AND
      ( P.pname = "milk" OR
        P.pname = "sugar") ;
```

c) Find the *cids* of companies who produce both "milk" and "sugar".

```
SELECT PR.cid
FROM PRODUCTS P, PRODUCE PR
WHERE P.pid = PR.pid AND
      P.pname = "milk" AND
      PR.cid IN      (SELECT PR2.cid
                      FROM PRODUCTS P2, PRODUCE PR2
                      WHERE P2.pid = PR2.pid AND
                        P2.pname = "sugar") ;
```

d) Find the *cnames* of companies who produce some products that are produced by the company having name "XYZ".

```

SELECT C.cname
FROM COMPANIES C, PRODUCE PR
WHERE C.cid = PR.cid AND
      C.cname != "XYZ" AND
      PR.pid IN      (SELECT PR2.pid
                      FROM COMPANIES C2, PRODUCE PR2
                      WHERE C2.cname = "XYZ" AND
                            C2.cid = PR2.cid);

```

g) Find the *cids* of companies who produce "milk" but not "sugar".

```

SELECT PR.cid
FROM PRODUCTS P, PRODUCE PR
WHERE P.pid = PR.pid AND
      P.pname = "milk" AND
      PR.cid NOT IN (SELECT PR2.cid
                    FROM PRODUCTS P2, PRODUCE PR2
                    WHERE P2.pname = "sugar" AND
                          P2.pid = PR2.pid);

```

h) Find the names of companies who do not produce any product.

```

SELECT C.cname
FROM COMPANIES C
WHERE C.cid NOT IN (SELECT PR.cid
                   FROM PRODUCE PR);

```

i) Find the names of companies who produce at least one product.

```

SELECT C.cname
FROM COMPANIES C, PRODUCE PR
WHERE C.cid = PR.cid;

```